

AI4Mobility

A multimodal assistive platform for parents supporting children with mobility difficulties. This deck covers the data: how it was constructed, cleaned, described and split.

PRESENTED BY

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What this deck covers

Four sections, ending with the numbers the pipeline makes possible.

1

The dataset

What it is, why it is synthetic, and what that does and does not license us to claim.

2

Cleaning

Four rules, 153 frames removed, and the deliberate decision to impute nothing.

3

Features

A four-stage pipeline and 45 features in three families.

4

Split, scaling and results

Subject-level split, a scaler fitted on training data only, and the model comparison.

5

Limitations and next steps

What the data layer cannot support, and what would fix it.

Team Deepminds — four roles, one platform

Four members, four defined responsibilities. This deck presents the data engineering work.

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THIS DECK

Data engineering and pipeline lead

Peer contribution split: 28%

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Model development and evaluation lead

Peer contribution split: 26%

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Pose estimation and visualisation lead

Peer contribution split: 24%

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Research and documentation lead

Peer contribution split: 22%

01

The dataset

1,358 raw frames from a seeded generator, six exercise classes, and an honest account of what that means.

The dataset is synthetic

It is not recorded video of children and it is not a public dataset. Every number describes the generator.

What this licenses us to claim

- The pipeline runs end to end and is reproducible under `random_state=42`.
- The models separate six classes that were constructed to differ as the real exercises differ.
- The evaluation protocol is sound and would transfer unchanged to real data.
- The confusion structure behaves as designed, which is evidence the pipeline measures what we intended.

What it does not license

- No claim about accuracy on real children performing real physiotherapy.
- No clinical claim of any kind.
- No claim that the platform improves adherence or technique.
- No comparison against published results on recorded datasets.

Why synthetic, and what is honest about it

- 1 Recording children needs approval we do not have**
Ethical approval, parental consent and safeguarding arrangements were out of reach for a taught project of this length.
- 2 No public dataset covers these six classes**
No paediatric physiotherapy pose dataset with our exercise classes was available to download.
- 3 The generator is seeded**
random_state=42, so every figure and every metric in the report reproduces exactly.
- 4 Class configurations come from the NHS descriptions**
Joint positions are derived from the published exercise text, so the classes differ in the ways the real exercises differ.
- 5 Degradations are injected on purpose**
Occlusion dropout, per-subject body proportions and measurement jitter, because a clean generator gives implausible accuracy.

The dataset in four numbers

1,358

Raw frames
Before any cleaning

153

Removed in cleaning
11.3% of the raw set

1,205

Usable frames
Carried into feature building

45

Features
Across three families plus quality

- Six classes: calf_control, cat_cow, child_s_pose, core_hip_stability, lower_limb, thread_the_needle.
- Requested frames per class range from 300 (lower_limb) down to 150 (calf_control).
- Annotation label noise of 6% was injected within confusable families.

FIGURE 2

Six classes, deliberately unbalanced

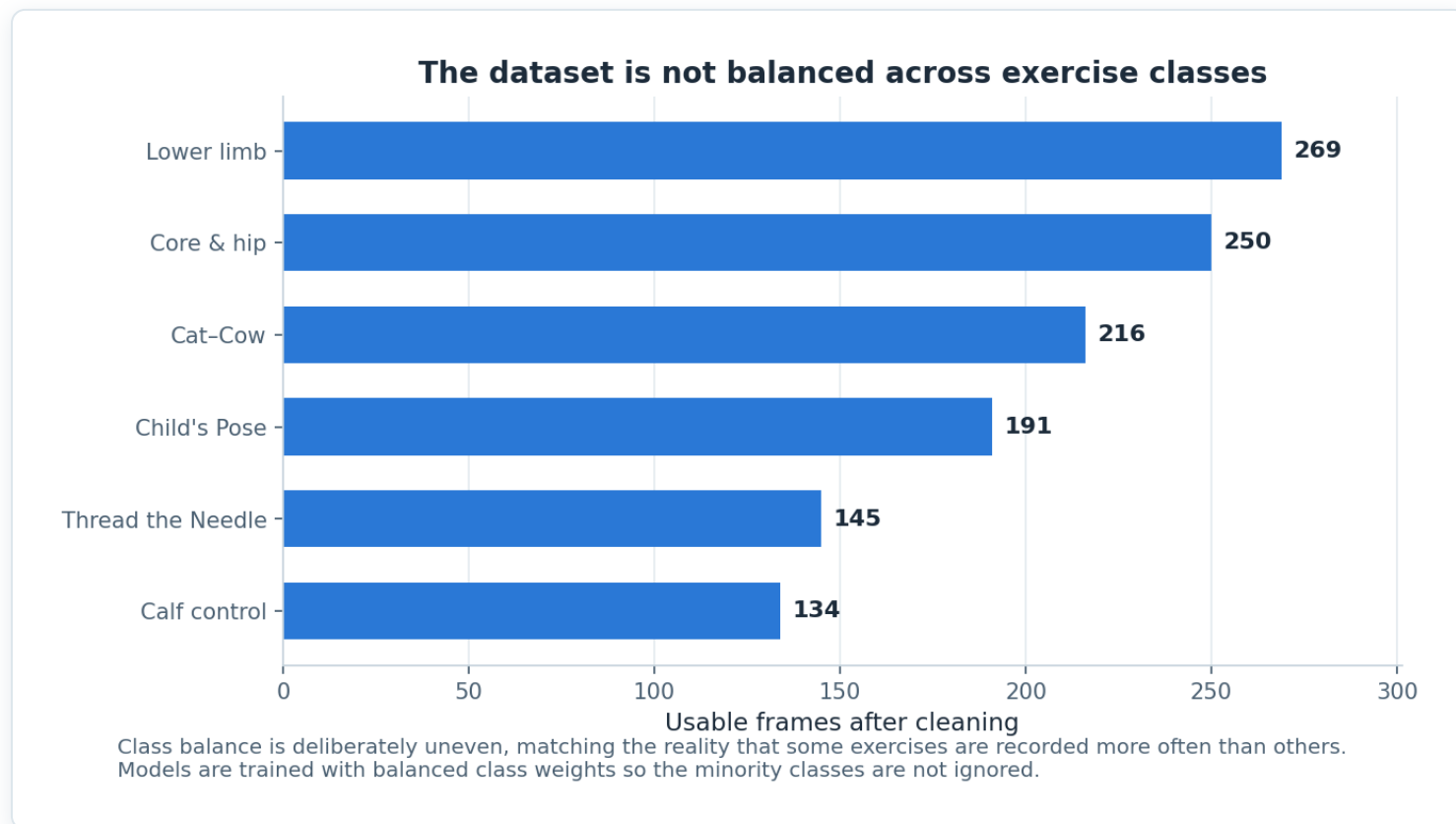


Figure 2 — usable frames per class after cleaning.

Why not balance them

- Requested counts: lower_limb 300, core_hip_stability 275, cat_cow 240, childs_pose 210, thread_the_needle 165, calf_control 150.
- Some exercises really are recorded more often than others, so an even set would be the unrealistic choice.
- Imbalance is handled in the model rather than in the data, using `class_weight="balanced"`.
- Macro F1 is reported alongside accuracy so small classes cannot be hidden by large ones.

02

Cleaning, and what we refused to do

Four rules, 153 frames removed, and no imputation anywhere in the pipeline.

Four rules, applied in order

Each rule is auditable and each removal is counted and reported.

Rule	What it removes	Frames removed
Exact duplicates	Identical rows, as when a recording is processed twice	18
Missing a critical joint	Any frame missing hip, neck, either shoulder or either knee	130
Detector confidence below 0.55	Frames the landmark detector was not confident about	5
Coordinates outside the frame	Landmarks outside the permitted bounds	0
Total	11.3% of 1,358 raw frames	153

Order matters: duplicates are removed first, so a duplicated bad frame is not counted twice.

WHY THE ZERO IS REPORTED

The out-of-frame rule removed nothing. It is still listed, because a check that finds no problems is evidence the data is well-formed, not a reason to hide the check.

FIGURE 3

What was removed, and why

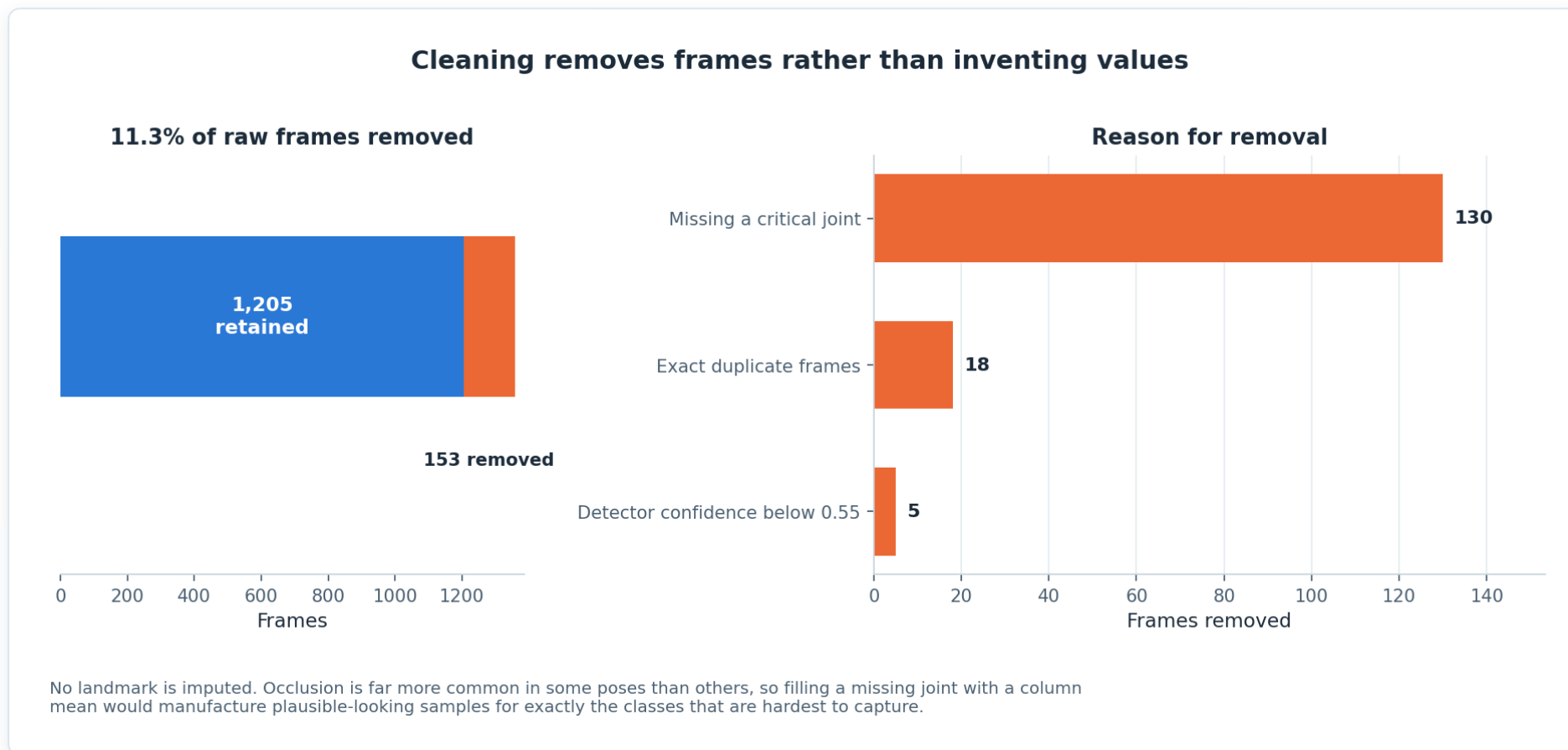


Figure 3 — 11.3% of raw frames removed, broken down by reason.

No landmark is imputed

Mean-filling would have been easy, faster and quietly wrong.

The reasoning

- Landmarks go missing through occlusion, not at random.
- Occlusion is pose-dependent: wrists and elbows disappear far more often in the quadruped poses.
- A column mean would therefore manufacture samples for exactly the hardest classes.
- Those invented samples would look plausible and would raise the reported score.

What we do instead

- Frames missing a critical joint are removed, and the removal is counted.
- Feature columns more than 35% empty are dropped rather than filled.
- Remaining rows with any missing value are dropped, so no value is ever invented.
- Detector confidence is kept as a feature, so frame quality stays visible to the model.

The pipeline runs in four stages

Each stage has one responsibility and hands a tabular artefact to the next.

1

Generate

`build_raw_frames` produces one row per frame from the seeded class-conditional generator. Replace this one function to swap in real data.

2

Clean

`clean_frames` applies the four rules and returns the surviving frames plus the removal counts.

3

Engineer features

`build_feature_table` produces 45 features in three families from 13 landmarks.

4

Split and scale

Subject-level train/test split, then a `StandardScaler` fitted on the training partition only.

THE POINT OF THE SEAM

Only the first stage knows the data is synthetic. Everything after it reads a plain landmark table, so a recorded dataset can be substituted without touching cleaning, features, splitting or modelling.

45 features in three families, plus a quality signal

Each family is included for a stated reason, not because it was available.

Family	Count	Why it is included
Normalised coordinates	26	Hip-centred and divided by torso length, so the model learns the pose rather than where the child stood or how tall they are
Joint angles	9	Invariant to translation, scale and body proportion — what an exercise classifier should key on
Inter-joint distances	9	Separations that distinguish classes, such as wrist-to-ankle for Child's Pose against Thread the Needle
Detector quality	1	Detector confidence, retained so frame quality stays visible instead of being silently averaged away

Frame-level features only. No temporal windowing, because the dataset is built as independent frames rather than continuous sequences.

WHERE THE 26 COMES FROM

Thirteen tracked landmarks, each with an x and a y, translated to a hip-centred origin and divided by torso length. Angles are returned as missing rather than zero when a segment is degenerate, so no invented value enters the table.

FIGURE 1

Thirteen landmarks, one representative frame per class

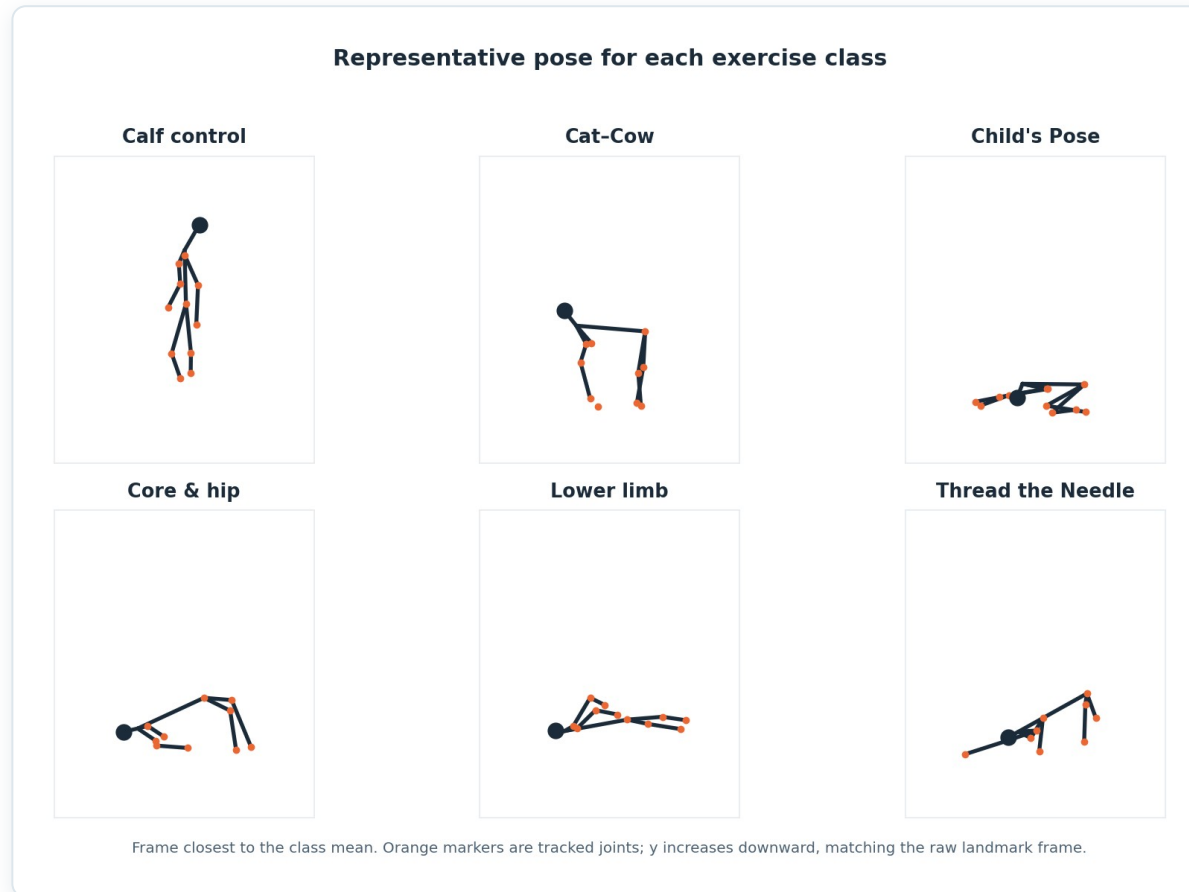


Figure 1 — the frame closest to each class mean, drawn with the shared skeleton connection list.

What the figure shows

- The pipeline tracks 13 landmarks: head, neck, shoulders, elbows, wrists, hip, knees and ankles.
- The representative frame is the one closest to the class mean, not the first frame — an unrepresentative frame makes a convincing and wrong figure.
- The four quadruped-derived classes look alike here, which is why the confusion structure later is not a surprise.
- y increases downward, matching the raw landmark frame.

FIGURE 4

Joint angles carry class structure

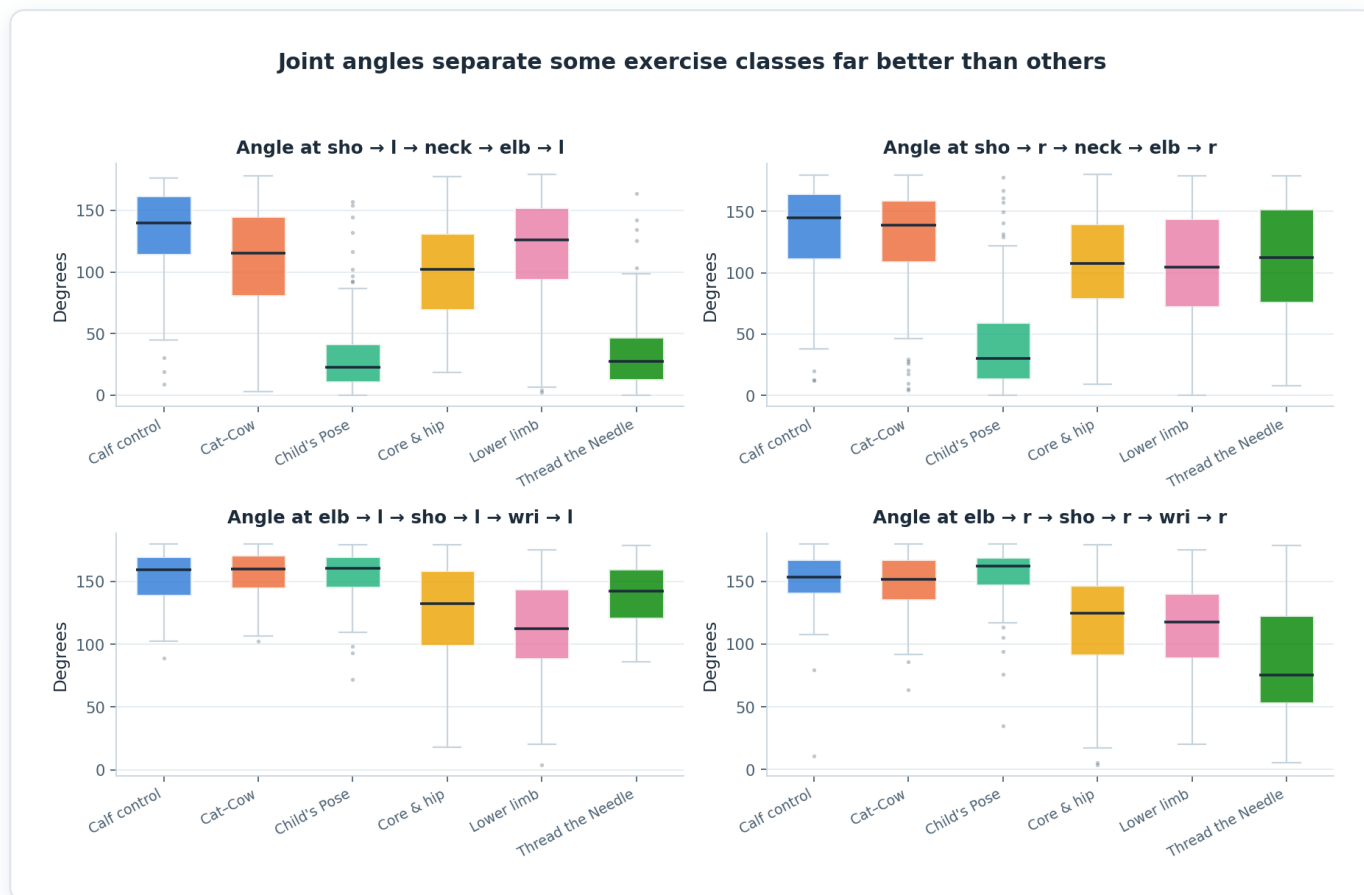


Figure 4 — joint-angle distributions by class, from the nine angle features.

Reading this honestly

- Nine angles are computed at anatomically meaningful vertices, such as hip-knee-ankle.
- Angles are returned as missing rather than zero when a segment is degenerate, so no meaningless value enters the table.
- Classes separate on some angles and overlap on others, which is the expected picture.
- Overlap between quadruped-derived classes is visible here and consistent with the later confusion analysis.

DO NOT OVERSTATE THIS

Angles help, but they do not dominate. Feature importance puts normalised positions at 77% and distances at 14%.

03

Splitting, scaling and results

The two decisions that decide whether a reported score means anything at all.

Split by subject, scale on training only

577 training frames and 195 test frames, partitioned by subject rather than by frame.

The subject-level split

- Frames are generated in short takes of up to 15, sharing one subject's body proportions.
- Whole subjects go to training or to test — never both.
- A random frame-level split would put near-duplicate frames on both sides.
- That silently inflates the test score, and nothing in the output would signal it.

The scaler

- StandardScaler is fitted on the training partition only.
- The fitted scaler is then applied to the test partition.
- Fitting before splitting leaks test means and variances into training.
- The leak is small, undetectable in the metrics, and would make every later comparison unsound.

THE HONEST VERSION

A frame-level split would have produced a higher number. It would also have been a measurement of memorisation rather than generalisation.

RESULTS

What this pipeline supports

Three model families, identical data and identical splits. Measured on synthetic data.

Model	Test accuracy	Test macro F1	CV macro F1 (5-fold, train only)	Train accuracy	Generalisation gap
Random Forest	0.928	0.935	0.955 ± 0.017	0.965	+0.037
Gradient Boosting	0.923	0.931	0.939 ± 0.016	1.000	+0.077
Neural Network (MLP)	0.923	0.932	0.939 ± 0.019	1.000	+0.077

Cross-validation is five-fold on the training partition only. The test set was scored once, after all model selection was complete.

READING THE TABLE

Random Forest wins narrowly on macro F1 and decisively on the generalisation gap: +0.037 against +0.077 for both alternatives. The gap is a pipeline property as much as a model property.

What the data layer cannot support

- **The data is synthetic**
No claim about real children follows from any figure in this deck.
- **Frame-level only**
There are no temporal features, so movement quality over time is not modelled at all.
- **Small test partition**
195 frames, and calf_control has a test support of only 14 — per-class figures for it are fragile.
- **Subjects are synthetic takes**
They stand in for people; they do not carry real between-person variation.
- **No clinical review of the class definitions**
The six classes come from published exercise text, not from a physiotherapist's judgement.

What would move the data layer forward

In priority order, and each one is achievable rather than aspirational.

1

Record a small consented dataset

Even 20 participants, with ethical approval and safeguarding in place, would let every number be re-derived from reality.

2

Add temporal features over a sliding window

Velocity and phase across frames, then re-run the same comparison on the same splits.

3

Keep the split discipline unchanged

A real dataset makes subject-level splitting more important, not less, since takes become recording sessions.

4

Report cleaning counts for real data too

The removal table becomes a data-quality instrument once frames come from a camera rather than a generator.

A pipeline built so the data can be replaced

Honest about the source

1,358 synthetic frames, seeded and reproducible, described as a generator rather than as evidence about children.

Disciplined in the middle

153 frames removed with reasons published, nothing imputed, 45 features in three justified families.

Trustworthy at the end

Subject-level split, scaler fitted on training only, test set scored once — 0.928 accuracy with a +0.037 gap.

The most useful thing this deck can leave behind is a method that survives contact with real data.

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